AMS 597 Statistical Computing
Spring 2026: Group Project Requirements
Instructor: Silvia Sharna

Group Selection: **(Due date 2/26/2026 by 5:00 PM)**

- Please form groups consisting of 4-5 members and select a group representative. This person will be the only medium of communication with me.
- Once you decide your group, **the group reresentative** will be required to email the instructor and copy the rest of the group **along with the Tas according to the below submission guidelines**.
- For those for whom finding a group is proving difficult, please email me latest by the above due date. I will randomly assign those individuals to a group.
- Email submissions from the group rep should include:
  - The names of your group members
  - The SBU IDs (in a pdf attachment).
- Late request(s) will be declined.
- Also, the group representative is required to submit all group information in Brightspace.
- Once you send the email and submit the group info in Brightspace, it becomes your FINAL group. No group modification is possible after that.

Dataset Criteria:

• Find a real dataset (preferably messy and dirty; so that you have the chance to show your data cleaning proficiency) which has $n \geq 100$ samples and $p \geq 20$ variables.
• Although your dataset contains $p \geq 20$ variables, not all variables need to be included in the analysis.
• The variables should include both categorical and continuous type.
• The dataset can be from any domain (e.g., healthcare, finance, social sciences, etc.).
• Provide a brief description of the dataset, including its source and relevance to the research question.

Confirmation of the Dataset: (Due date 3/5/2026 by 5:00 PM)

- Once the dataset is finalized, the **group representative** will be required to email the dataset to the instructor and copy the rest of the group **along with the TAs**.
- The group representative is required to submit all dataset information in Brightspace.
- Your submission should include:
  1. The dataset.
  2. If the dataset is too large, please paste the link of the dataset/Google drive link in a pdf file and submit that.

- The **group representative** will be responsible for submitting the required materials in Brightspace upon completion of the project. Hence, choose your representative wisely!
    - This group representative will be required to submit all the project materials on behalf of the group.
    - Other group members will be required to submit the report (pdf file) only.
- Once I get the dataset info along with the group info from the representative, I will assign a group number for each group (first come first served!).

Data Preprocessing:

- Perform necessary data cleaning (e.g., handling missing values, outliers, etc.).
- Provide a brief justification of any data cleaning process that you have performed (if any).
- Conduct exploratory data analysis (EDA) using advanced visualization techniques (e.g., ggplot2, plotly, etc.).
- Interpret trends, patterns, and relationships between variables.
- Transform variables if needed (e.g., scaling, encoding categorical variables, etc.).

Statistical Analysis:

You are required to analyze your data in R (RMarkdown) or Python (Jupyter Notebook) only. Your project should include the following components:

1. Introduction & Objective

- Briefly describe the dataset and explain the goal(s) of your analysis.
- The research should be addressing Three different questions (but can be related) of interest. That is, it SHOULD NOT be using two different methods to answer the same scientific question.
- For one of the research questions, you are allowed to use Python; but for the rest of the two you are required to analyze your dataset using R only.

2. Modeling or Statistical Inference

- The first research question can be answered using any of the methods that we are learning in class or more advanced methods.
- The rest of the two research questions should use two different advanced statistical methods such as:
  o Principal Component Analysis (PCA) or Factor Analysis.
  o Clustering (e.g., k-means, hierarchical clustering).
  o Resampling method.
  o GLM

o Analysis of multi factor experiments
o Mixed effect model
o Regression modeling (e.g., linear regression, logistic regression, Lasso, Ridge).
o Time series analysis (if applicable).
o Machine learning techniques (e.g., decision trees, random forests, etc.).
o Justify the choice of methods based on the dataset and research question.

3. Model Evaluation:

- Evaluate the performance of your models using appropriate metrics (e.g., RMSE, AUC, accuracy, etc.).
- Interpret the results and discuss their implications.

Reproducibility and Code Quality:

- Write clean, well-documented R code using functions, loops, and vectorized operations where appropriate.
- Use only R Markdown (for python codes, Jupyter notebook) to create a reproducible report that includes:
  o Code chunks.
  o Visualizations.
  o Interpretations of results.
- Ensure the code is well-organized and easy to follow.
- Your code should include comments describing the purpose of each chunk of code (i.e., what each chunk of the code is trying to analyze and should be clear from the comments included).

Report Writing:

- Write a report (maximum 25 pages excluding appendices (shorter is fine), font size ≥ 11).
- You can use Word/LaTeX/Markdown to write the report.
- The report should be self-contained. Submit your report as a pdf file. Be sure to include:
  o Introduction and research question.
  o Description of the dataset.
  o Methodology and implementation details.
  o Results and discussion.
  o Conclusion and limitations.

Submission Guidelines:

- You must submit the dataset along with the source code (as a .csv). If the data is large, use SBU google drive to upload the data.
- Submit your source file (i.e., Rmd file and/or ipynb file) with well-commented, organized code together with the pdf.
    - Your notebook should be **readable, reproducible, and executable** from top to bottom.
    - Include markdown cells for explanations, interpretations, and transitions between sections.

Presentation Slides:

- Each group will also prepare and submit PPT presentation slides (maximum 15 slides) summarizing your project.
- The PPT slides might be shared with the rest of the class.

Submission Deadline (Read this section carefully!): (Due date 4/22/2026 by 5:00 PM)

- All materials MUST be submitted by no later than April 22, 2026 by **5:00 PM**.
- ONLY the group representative will submit all the project materials (dataset, report, presentation slides, R/Python code) **in the Brightspace, and email to the instructor (silvia.sharna@stonybrook.edu) and cc the TAs and the other group members**.
- Although you work in a group, **each student is still expected to submit the report (only the pdf) on Brightspace**. If you do not submit the report on Brightspace, there will be a penalty on your project score.
- In the Subject line of the email, type "AMS 597 Spring 2026 Group XX Project", where XX will be replaced by your assigned group number.

Presentation:

• Presentations will start on 4/22/2026 with the hope that we will be able to wrap up everything by 5/8/2026.
o But in case we are not able to, then we will be meeting on the Final Exam day which is Wednesday, **May 20 from** 11:15am 1:45 pm.
• Each group should prepare a 20-minute presentation.
• We have 75 students in total. The number of groups will be between 15 to 19.
• We plan to do 6 presentations each day, but the 7th group should always be ready in case we can accommodate them within the duration of the assigned class time.
• Be prepared to answer questions from the instructor, TAs and peers.

Some Potentially Useful Data Archives/Repositories:

You are encouraged to find your own real dataset. If you have trouble finding your own dataset, you can check the publicly available datasets from the following repositories:

• Data and Story Library (DASL): http://lib.stat.cmu.edu/DASL/
• NIST Statistical Reference Datasets: http://www.itl.nist.gov/div898/strd/

- UCI Machine Learning Repository: http://archive.ics.uci.edu/ml/
- UMASS Datasets: http://www.umass.edu/statdata/statdata/data/
- DataSF: https://datasf.org/
- Sports Statistics: https://sports-statistics.com/sports-data/
- Other Data Repositories: https://www.kdnuggets.com/datasets/index.html

Grading Criteria:

- Are the report and presentation slides well-organized?
- Is supporting computer output provided (in edited form, that is, edit out all the extraneous information in the report)? R (and/or Python if you decide to use it to address one research question) must be used for model fitting and plotting.
- Is the model appropriate for the design and questions of interest? Have you checked the assumptions?
- Are correct interpretations given for the parameters in the model?
- Are the conclusions drawn from the model correct and do they answer the question of interest?
- Are the research questions **interesting** and non-trivial?
- Can the instructor reproduce the results reported using the code the group provided if the model is correct?

Group Synergy:

- If you have concerns with non-contributing members and are not able to resolve them within the group, please speak to the instructor immediately (Please DO NOT wait until project due date).
- There will be an optional peer evaluation for groups with potential group synergy issues:
    - For the groups with non-contributing members, for each member of the group, fill in the peer evaluation on a scale of 0-100%, how much each of the other group members should get from the group project score (e.g., if the group gets 24/30 and member A is given 60% by member B, 50% by member C, 80% by member D and 70% by member E and member A evaluates his/her own-self 100%, the final score for member A will be 17.28/30). The instructor will arrange for zoom meetings with these groups to ensure fair evaluation.
      *****Note: Plagiarism or the use of pre-existing analyses without proper attribution will result in a failing grade. Be creative and demonstrate your understanding of statistical computing in R.**